

Project Report: E-Commerce Order Microservice - Java Backend Developer

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Family:	Java Backend Developer
Level:	Intermediate
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Order service with inventory checks, JWT auth, Redis cache, and Kafka event publishing.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Intermediate]

Chapter focus: Project Overview** *(Level: Intermediate)

E-Commerce Order Microservice** is a intermediate-level end-to-end project for Java Backend Developer. Duration: 4-5 weeks. Order service with inventory checks, JWT auth, Redis cache, and Kafka event publishing. Tools: Spring Boot 3, Kafka, Redis, PostgreSQL.

Code Reference:

```
Objective: Implement JWT authentication
Objective: Publish OrderCreated events to Kafka
Objective: Add Redis caching for product catalog
Objective: Integration tests with Testcontainers
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Java Backend Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Java Backend Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Domain [Intermediate]

Chapter focus: Phase: Domain** *(Level: Intermediate)

Order and OrderItem entities with state machine. Complete these tasks: Order statuses; Inventory client. Deliverable for this phase: Domain model docs.

Code Reference:

- [] Order statuses
- [] Inventory client

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Domain is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Domain model docs to the teacher.

Chapter 3: Phase: Security [Intermediate]

Chapter focus: Phase: Security** *(Level: Intermediate)

OAuth2 resource server. Complete these tasks: JWT validation; Role-based endpoints. Deliverable for this phase: Secured API.

Code Reference:

- [] JWT validation
- [] Role-based endpoints

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Security is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Secured API to the teacher.

Chapter 4: Phase: Events [Intermediate]**Chapter focus: Phase: Events** *(Level: Intermediate)**

Kafka producer/consumer. Complete these tasks: OrderCreated topic; Inventory listener.
Deliverable for this phase: Event flow diagram.

Code Reference:

- [] OrderCreated topic
- [] Inventory listener

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Events is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Event flow diagram to the teacher.

Chapter 5: Phase: Observability [Intermediate]**Chapter focus: Phase: Observability** *(Level: Intermediate)**

Metrics and structured logging. Complete these tasks: Micrometer; JSON logs. Deliverable for this phase: Grafana dashboard screenshot.

Code Reference:

- [] Micrometer
- [] JSON logs

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Observability is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Java Backend Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Grafana dashboard screenshot to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Intermediate]

Chapter focus: Final Deliverables and Review *(Level: Intermediate)**

Submit all final deliverables: Architecture diagram; Running microservice; Grafana dashboard; Postman collection. Skills demonstrated: Microservices, Kafka, Security, Caching. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: Architecture diagram
Deliverable: Running microservice
Deliverable: Grafana dashboard
Deliverable: Postman collection
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits Architecture diagram and passes the rubric.